

AWI-ESM3 ↔ PISM moving-cavity coupling

the investigation report

Single living document. Superseded claims are struck through, not deleted.

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Abstract

Mission. A working end-to-end AWI-ESM3 (FESOM2–CORE3-cavity + OIFS-48r1 + LPJ-GUESS + WAM) ↔ Antarctic-PISM cycle: awiesm3 year → PISM decade → awiesm3 year on a changed mesh, with all three coastline-following members (FESOM cavity, OIFS land–sea mask, WAM sea points) regenerated per leg.

Where we are. Five real bugs have been found and fixed (§1); the wave-model coastline is solved structurally (§1.5); the workflow performs genuine awiesm3→PISM→awiesm3 handovers (§1.7); and the recurring OIFS crash family was traced past three partial fixes to its true root: **the ice–atmosphere surface-temperature coupling architecture itself is unstable in the marginal ice zone** (§1.6). The atmosphere’s ice-tile skin was hard-pinned to the remapped FESOM ice temperature (skin conductivity 10^{10}), so kW-scale spurious fluxes circulated between the two models. The fix is architectural: OIFS now solves its own ice-tile skin implicitly each timestep, with slab conduction through the *coupled* FESOM ice thickness (new exchange field) and snow — the “coupled slab” (§2) — validated: the first slab run completed its year crash-free with the kilowatt flux events collapsed to zero.

Remaining open items: the increment test itself — the chunk-3 leg on the changed mesh dies after weeks of model time from **phantom zero-water ingestion at draft-changed cavity columns** (mechanism proven, fix candidates identified, §3) — and the too-weak Southern-Hemisphere summer sea-ice melt-back relative to the 500-year v3.4 reference (§4).

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1 The campaign, in the order it happened

Each subsection carries the date it was worked. The bug numbering (I–V) is the order of *discovery* and is kept as the canonical naming.

1.1 The restart seam and the cold-start control (2026-07-08–13)

The original FESOM cavity-margin blowup: a remapped warm restart mixes an evolved, balanced dynamic state (98% of nodes, bit-exact) with patched values at changed nodes; the *seam* between them is the instability. The decisive control:

Run	Δt	Initial state	Outcome
full-mesh control	1200 s	native spun-up restart	stable, worst CFLz 2.61
PISM cavity, cold	1200 s	cold start	stable, CFLz 3.44
PISM cavity, warm	1200 s	remapped restart	dies day 4.0 (CFLz 9.2)
PISM cavity, warm	60 s	remapped restart	dies day 4.4 (η_n , 0 CFLz)

The ocean lives happily in PISM’s cavity; the production answer became: run chunk-1 cold-started on the PISM cavity (artifacts pooled), and let `remap_restart` carry only genuine per-leg increments. The first real increment test crashed on 17 fully-opened columns (ice removed entirely, 16 levels at once) out of 1399 — but its numbers are **TAINTED** (measured under Bugs I+II) and the finding that matters survived into the fixes below: absorbing 99% of a geometry change is not a passing grade when the remaining 1% is fatal.

1.2 Bug I — PISM was fed surface water as ice-base forcing (2026-07-14)

`coupling_ocean2pism` built PISM’s `-ocean` th forcing as a mean over FESOM level *indices* 1–20 = **0–410 m including the summer surface** (the correct `-sellevel,150/600` sat in dead code). PISM received ice-base water up to +6.5°C even from a healthy ocean, melted at ~30 m/yr, and shed **24% of its floating area per decade** — every mesh-change leg then had to absorb a catastrophic geometry increment. FESOM’s own cavity melt in the same runs was sane (2.2 m/yr mean, 0.72 median).

Fix (93a207ee): the DIRECT route. PISM consumes FESOM’s own cavity fluxes via `-ocean` given: `fw→shelfbmassflux` ($\times 1000$, sign preserved), `sst→shelfbtemp`. One model owns the ice–ocean interface. The `-ocean` given machinery existed all along but silently fell back to `th` because its input is a FESOM1.x-era bundle FESOM2 never wrote; `fesom2ice` now synthesizes it (plus three exhumations of never-run 2D-path code: 3f07d7a2, 74671c27, bb9d00c2).

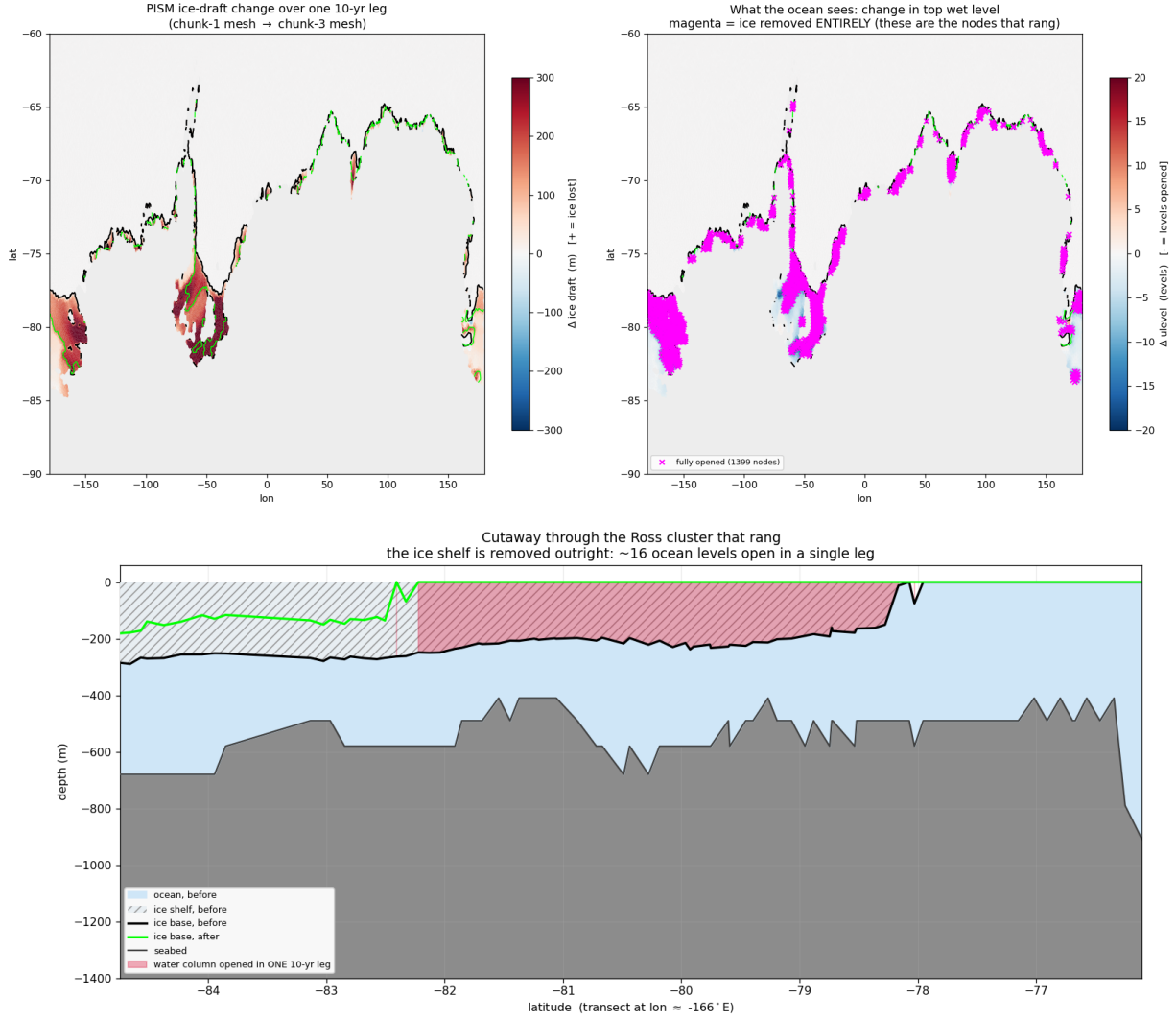


Figure 1: What Bug I did to PISM in one 10-year leg under surface-water forcing: 1399 nodes lose their ice entirely (magenta ring around the whole margin), the Ross front retreats ≈ 440 km (section: ice base $-200 \dots -280$ m $\rightarrow$ open ocean across four degrees of latitude).

1.3 Bug II — the exchange weights scrambled the coupled ocean (2026-07-14)

FESOM exchanges coupling fields in *partition* (*my_list/rank*) order; the offline weight engine (2026-06-17) built the feom geometry in *nod2d.out* order (median displacement 58°). Every submesh exchange was geographically tile-translocated (Fig. 2): OIFS saw Weddell ice fraction 0.04 while FESOM had 0.95; global sea-ice death followed within months.

Fix (c21fe7c2): `permute_feom_to_runtime.py` reorders the OASIS geometry to runtime order (via `dist_nproc>rpart.out`); `validate_feom_weights.py` applies every weight file to real latitudes and **aborts the leg** above 1° median error (nod2d-ordered weights fail at $42\text{--}58^\circ$, correct ones pass at $\leq 0.025^\circ$).

1.4 Bug III — the OASIS coupling restart was compound-scrambled (2026-07-14)

`remap_oasis_restart.py` gathered in nod2d space, but its input and required output are both *run-time*-ordered: the emitted `rstos.nc` matched *neither* ordering (correlation of `sst_feom` with the restart's own surface temperature: $+0.15$). OIFS's first coupling interval saw 305 K water under po-

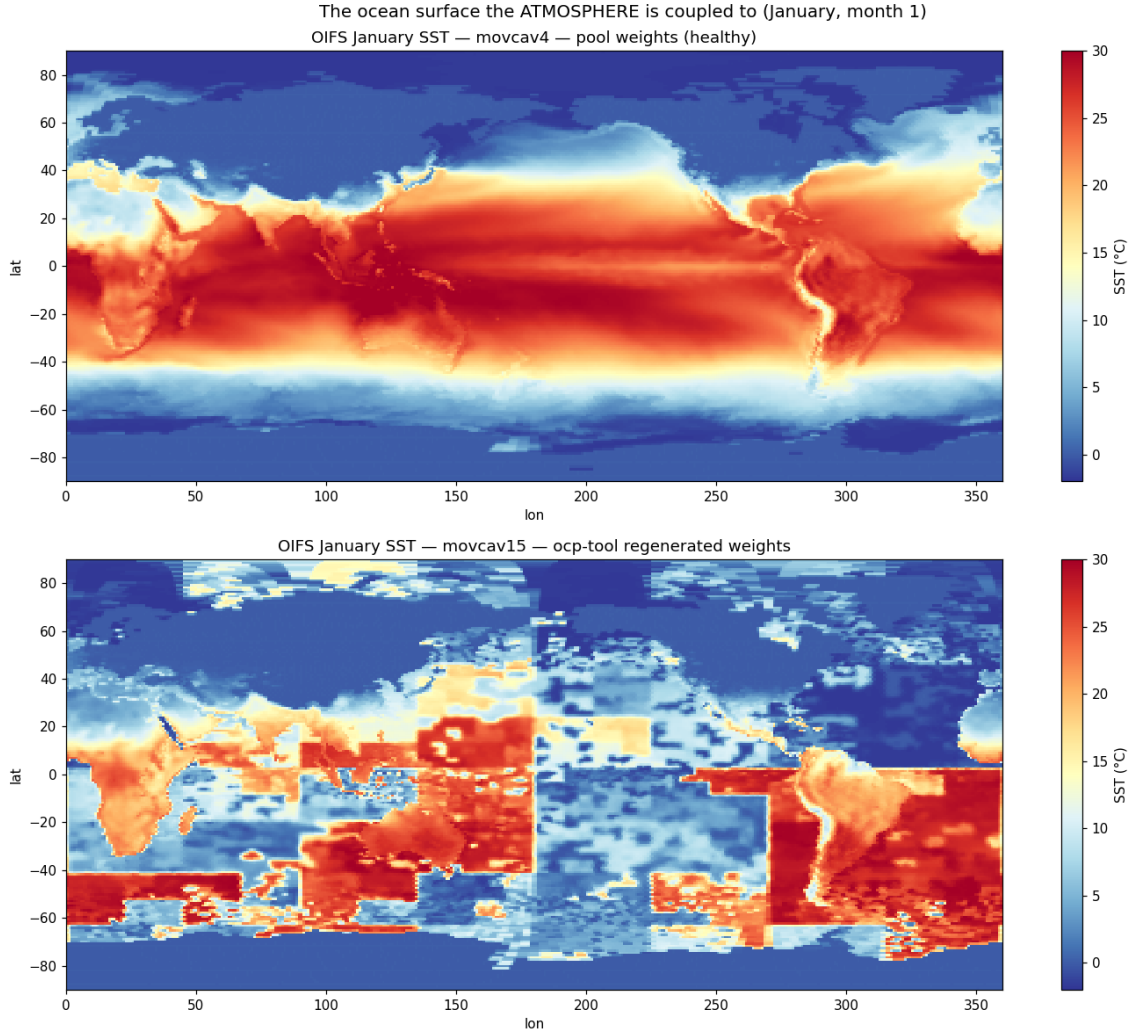


Figure 2: **The picture that settles it.** January SST as OIFS holds it. Top: pool weights — textbook. Bottom: offline-engine weights — a *tile-translocated jigsaw*: tropical blocks at 55°S, polar blocks on the equator, a seam at lon 180 where the node ordering breaks.

lar air — a one-hour global jigsaw shock that nondeterministically detonated the convection scheme (identical inputs crashed at hour 19, hour 458, or never).

Fix (abe56ee2): un-permute via the old mesh’s `rpart.out` → gather → re-permute via the new mesh’s; correlation now +0.995. Pool `rstos.nc` rebuilt.

1.5 The wave model’s frozen coastline (2026-07-15)

Chunk-3 died deterministically at its first physics step in `voskin_mod.F90:282` squaring WAM’s Stokes drift. WAM has no runtime tiling: its sea-point set is baked into `wam_grid_tables` at preprocessing time and had never been regenerated. A cell that is ocean per the regenerated lsm but land per WAM never receives wave data; the Stokes/Charnock fields are srf-restart-carried, so a *warm* start after a coastline change reads never-written storage (cold starts zero-initialize — why chunk-1 tolerates its 12 mismatch cells and historical lsm changes never hit this). Chunk-3 had 125 mismatch cells. WAM itself was separately exonerated for the Bug-IV crash family (a WAM-off run crashed identically).

Fix: one-time maximal-ocean tables (option A; tolerance patching rejected). WAM’s sea-point set was rebuilt from ETOPO1 with the southern limit extended 78 → 85°S, then *max-ocean edited*: every cell the FESOM max-mesh can make wet — open ocean or sub-shelf cavity — is forced

to sea at its FESOM bottom depth (only ever adding ocean). 1277 cells opened (29 555 $\rightarrow$ 30 832 sea points), 283 of them south of 55°S: exactly the band a moving coastline walks through (Fig. 3). The tables never change again, wave restarts stay index-compatible across all future mesh changes, and WAM cold-starts once at the switch leg. Known cosmetic debt: the max-ocean edit did not update the obstruction coefficients.

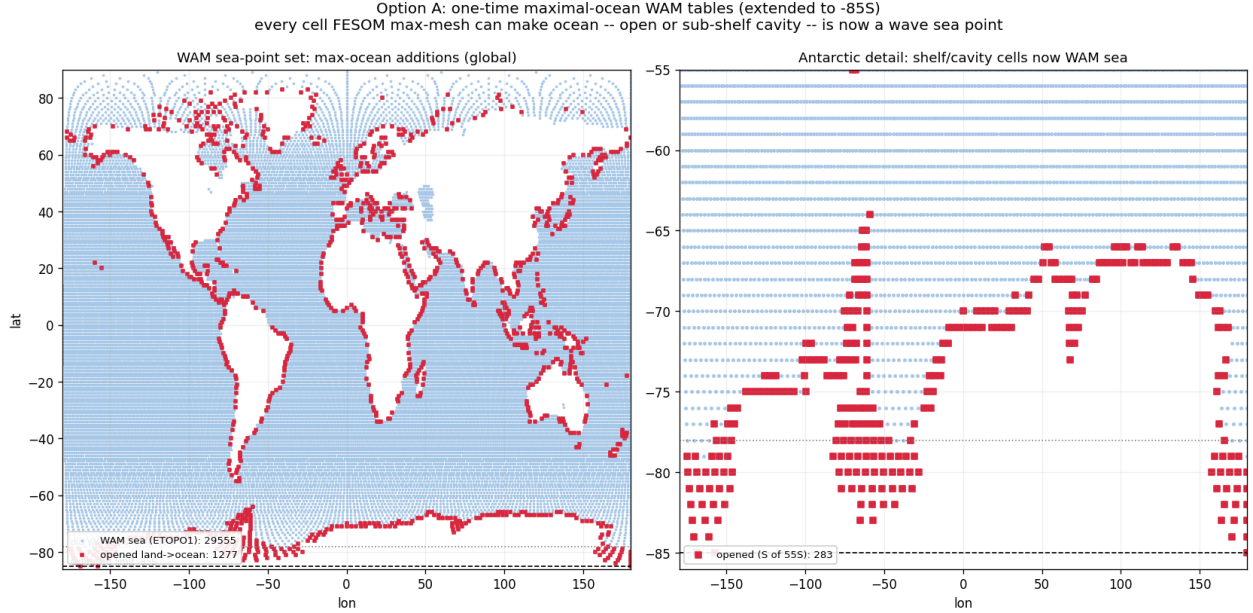


Figure 3: One-time maximal-ocean WAM tables (extended to 85°S). Blue: wave sea points from ETOPO1. Red: cells opened land $\rightarrow$ ocean because the FESOM max-mesh has a water column there. Pre-registering them makes the `voskin` step-0 crash structurally impossible on any future leg.

1.6 Bug IV — the ice-surface-temperature coupling is architecturally unstable (2026-07-14 $\rightarrow$ 16)

This is the mid-run OIFS crash family (`cubasen/vsurf/voskin` floating overflow, random timing, polar/MIZ ranks). It survived three partial fixes before the root was found; the full chain is worth recording because each layer was real, just not sufficient.

Layer 1: MIZ aliasing of the sent temperature. FESOM sent raw `ist`; ice-free nodes send the seawater freezing point, so the GAUSWGT cell blend in the marginal ice zone was dominated by warm open water: OIFS’s ice tile saw a surface tens of K too warm and returned a strongly negative flux to the fully-iced nodes. Partial fix: concentration-weighted `ist` (`ist.a_ice` $\div$ `ci` on ingest, the EC-Earth/NEMO convention behind `ECE_CPL_NEMO_WEIGHTED_ICE`). Runs survived $3\times$ longer, still crashed.

Layer 2: the FESIM solve has no flux feedback within the coupling interval. FESIM’s skin solve relaxes toward $t^* = T_f + a2ihf \cdot zsniced/con$ with the received flux *frozen* for the hour and no dQ/dT term: under insulation the equilibrium runs away (measured floors 145–183 K; excursions to -954 K in one run). Partial fix: an implicit flux linearization anchored at the transmitted temperature, $Q(t) = a2ihf + \lambda (t_{ref} - t)$ with $\lambda = 4\epsilon\sigma t_{ref}^3 + \lambda_{turb} \approx 20$ W/m²/K (the NEMO-family fallback derivative), energy-closed by booking the linearization heat to the ice growth budget (fesom branch `awiesm3-implicit-ice-surftemp`, 38a558b5). Dips became bounded and self-recovering in most runs — but not all.

Layer 3 (the root): the atmosphere’s ice skin was never solved at all. Send-side instrumentation of `A_Q_ice` (term decomposition + tile state at every $|Q| > 1 \text{ kW/m}^2$ event) delivered the verdict:

- The received flux at cold FESOM nodes is **unphysical and anti-braking**: mean -15.8 kW/m^2 below 150 K (100% negative), -4.5 kW at 150–180 K, recurring all year with the MIZ seasonal cycle — at **49°N on 0.1–0.5 m ice with thin snow** (Okhotsk / St. Lawrence). Insulation is irrelevant there; no realistic λ brakes kilowatts.
- At the OIFS send, the generating cells have ice-tile fractions down to 0.04 and **tile skin temperatures of $\sim 204 \text{ K}$ under $\sim 275 \text{ K}$ maritime air**; the sensible + latent terms then reach $+17.8 \text{ kW/m}^2$ per tile area. With weighted-ist the tile skins were *still* 189–204 K at substantial tiles — the cold temperature is genuine FESIM state, faithfully communicated.
- The reason the skin can sit 80 K below the air: `vlamsk_mod.F90`, sea-ice tile, with `LNEMOSICOUPL=.true.` (the default): `PLAMSK = 1e10` — **the skin conductivity is set to infinity, pinning the tile skin to the prescribed (remapped FESOM) ice temperature**. The atmosphere’s surface scheme was forbidden from solving its ice surface; it computed fluxes against whatever arrived through the remap.

The loop, in one line: FESIM’s oscillating solve $\rightarrow$ remap $\rightarrow$ *pinned* OIFS skin $\rightarrow$ kW fluxes against an unresolvable surface $\rightarrow$ remap $\rightarrow$ FESIM slams further — a two-model positive feedback closed through the marginal ice zone, metastable enough that identical configurations coin-flipped between completing a year and dying at day 2 (runs `movcav25–31`: `raw/weighted` $\times \lambda = 3.4/20 \times$ `dirty/clean` initial state — every combination crash-prone).

Root fix: the coupled slab (§2). Stop prescribing the atmosphere’s ice skin altogether; let OIFS solve it implicitly each timestep with conduction through the *coupled* FESOM ice and snow thickness. This removes the loop by construction while keeping the flux honest about the actual ice state. Upstream note: mainline AWI-CM3 carries the same pinned-skin architecture and therefore the same latent (if rarer) crash mode.

1.7 Bug V — iterative coupling never handed over to PISM (2026-07-16)

The `awiesm3` $\rightarrow$ `PISM` model switch is resolved at `SimulationSetup` init by reading the `chunk_date` file, whose path `esm_runscripts/chunky_parts.py` builds by *raw string concatenation of `general.base_dir` before `esm_parser` resolves variables*. A `base_dir` containing `#{user}` was probed verbatim; the silent `isfile()` miss fell back to “chunk 1, model 1” — `awiesm3` self-resubmitted to its own final date and `PISM` never ran. Proven by log comparison: identical resubmission command lines resolve to `Valid Model Names = ['pism']` under a literal `base_dir` (`movcav16` era) and to `['fesom', 'oifs', ...]` under `#{user}`.

Fix (d61297e8): `_early_resolve()` expands `#{...}` against `general` (environment fallback) in all three chunk-file path builders. Regression-tested against a real chunk file, and validated live: the first genuine `awiesm3` $\rightarrow$ `PISM` handover of the campaign (`movcav27`), and a full `awiesm3` $\rightarrow$ `PISM` $\rightarrow$ `couple_out` $\rightarrow$ `chunk-3-couple_in` chain (`movcav29`).

2 The coupled-slab architecture (2026-07-16/17)

The new division of labour: **OIFS owns the ice surface temperature; FESOM owns the ice state**. OIFS’s surface scheme solves the ice-tile skin implicitly within every atmosphere timestep (full radiative + turbulent dQ/dT , finite skin conductivity), with slab conduction through the coupled per-ice thickness and snow depth. FESOM’s albedo still drives the radiation; FESIM’s own `ist` solve remains for its internal state (albedo, melt ponds) but is read by nobody — the feedback loop is cut

by construction. The machinery is EC-Earth/KNMI heritage (ICESTATENEMO, 2008; the SI3 branch of `callpar`) — our contribution is wiring the FESOM branch, which lacked only the thickness.

New exchange field. FESOM sends `sit_feom` = effective (grid-mean) ice thickness `m_ice` (o2a collection, `A_Ice_thickness`); `ECE_FESIM_GET_ICE_STATE` divides by the received ice fraction (standard ε -guard) for the per-ice thickness the slab needs, falling back to the climatological 1.5 m only where the cell has no significant ice. Snow thickness (`snt_feom`) was already exchanged and now actually reaches `surfseb` via `SURFL%ZSNTICE`.

The switch set (verified in code, not inferred):

switch	setting	verified effect
LNEMOSICOU	.false.	<code>vlamsk</code> : sea-ice tile skin conductivity becomes <i>finite</i> (land-tile-style lookup) instead of 10^{10} — the skin is genuinely solved by the SEB. The prior default pinned it to the prescribed ice temperature; this pin was the kilowatt engine of Bug IV.
LNEMOLIMTEMP	.false.	disables the hourly overwrite of OIFS’s 4-layer ice temperature (<code>SP_SB YTL</code>) from the FESOM ist ingest; the profile is OIFS-prognostic (ICMGG-initialized, restart-carried). FESOM’s ist send becomes inert.
LNEMOLIMTHK	.true.	activates the <code>callpar</code> fill of <code>ZTHKICE/ZSNTICE</code> from the coupled fields — the skin/top-layer conduction feels the real ice thickness and snow insulation. (The deep 4-layer discretization keeps its fixed depths; the skin conduction is the term that controls winter fluxes.)
LMCCDYNSEAICE	.true.	master enable; false force-resets the whole LNEMOLIM* family (<code>sumcc</code>).
LNEMOLIMALB	.true. (default)	FESOM’s albedo drives OIFS radiation (<code>surfrad_layer</code> → FESIM getter; raw–raw convention, consistent).
ECE_CPL_NEMO_WEIGHTED_ICE	.true. (inert)	only acted inside the disabled ist ingest.
FESOM latm_owns_ist	.true.	FESIM growth budget takes the atmosphere flux directly (<code>Qatmice=-a2ihf</code> , the ECHAM convention); <code>qres/qcon/qlam</code> booking off.

The `sumcc` guard “LNEMOSICOU without LNEMOLIMTEMP does not make sense” correctly enforces that a pinned skin requires a prescribed temperature; our mode flips both off. In one sentence: *before, OIFS’s ice surface was a mirror bolted to whatever FESOM’s oscillating solve sent; now it is a real solved surface conducting through FESOM’s actual ice and snow.*

Validated (movcav33). The first coupled-slab run completed its year crash-free and handed over to PISM. With the instrumentation still armed: kW-scale send events at real ice tiles went from 1410–1930 per run to **zero** (117 residual events are all phantom slivers with ice fraction $\sim 10^{-7}$, a `ZMASK` threshold cosmetic); event tile skins moved from 189–212 K into the physical 247–265 K range (Fig. 4); the fluxes FESOM receives at its coldest nodes are $-63 \dots -80 \text{ W/m}^2$ (textbook polar values, was -10 kW), and FESIM’s internal temperature tracks its anchor to $< 0.1 \text{ K}$.

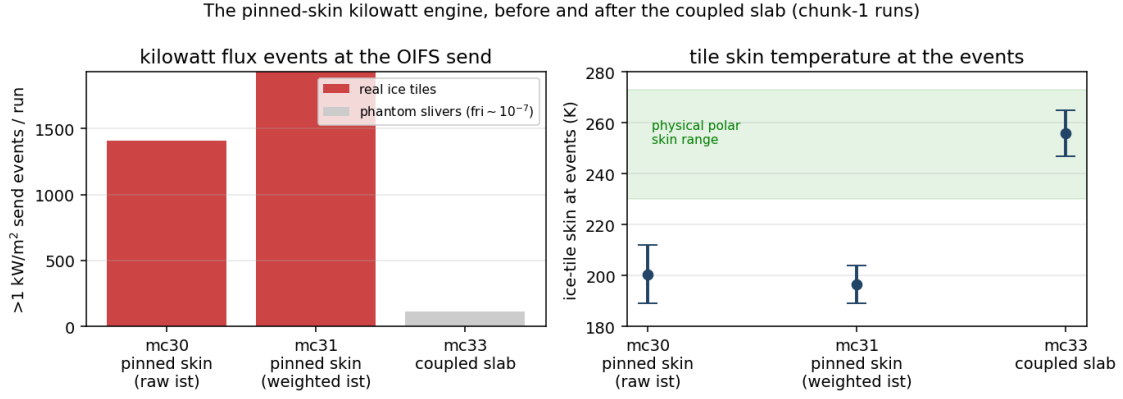


Figure 4: The kilowatt engine, before and after. Left: $>1\text{ kW/m}^2$ events at the OIFS send per chunk-1 run. Right: the ice-tile skin temperature at those events — pinned-skin runs sit 60–80 K below the physical range; the solved skin cannot.

3 The increment test crashes: phantom zero-water at changed cavity columns (2026-07-17)

The first chunk-3 leg — awiesm3 on the PISM-changed 208 438-node mesh, stage 6/6 — absorbs the mesh change and runs for weeks of model time before FESOM dies of an SSH explosion (η jumping $\pm 40\text{ m}$ in one step at the Antarctic Peninsula tip; day 19–51 depending on MPI reduction order — the state is metastable). The autopsy was made cheap by two facts: the leg costs 3–4 minutes of wall time, and a temporary XIOS file group with `sync_freq="1d"` flushes daily 2-D/3-D fields to disk as they complete, so even a crashed run leaves full evidence (the default XIOS files sync only on close: 51 days of output, zero records).

Mechanism. Cavity columns have a pinned free surface ($\eta \equiv 0$, layer thicknesses frozen), so nothing relaxes a flow imbalance there. The remap leaves a small divergence at draft-changed columns that drives vertical velocity through the ice base (seeded at 10^{-4} m s^{-1} , exactly the observed early stripping rate; it grows $100\times$ through a feedback — stripping makes the column buoyant, buoyancy drives more w — visible as 78 704 CFL_z warnings in 51 days vs. 125 per *year* in chunk-1, every hotspot within 0.1° of a changed column). FESOM’s vertical tracer machinery treats the top interface like a *free surface*, where entering water carries zero tracer (correct for rain; `oce_adv_tra_ver.F90`, top-layer coefficients) — under an ice shelf the same treatment strips T and S toward $(0,0)$ for as long as w persists. Chunk-1 is immune because balanced cavities have $w(\text{base}) \approx 0$; the flaw is armed only by a mesh change.

Ruled out by measurement (each was a plausible suspect): surface freshwater fluxes (`fw` is $50\times$ too small at the injectors, and *negative* over the ring); OIFS runoff/calving and the `ECE_LANDICE` snow cap (mapper config identical between chunks, remap weight sums ≤ 1 , ICMGG sd hygiene clean); the OASIS `feom` masks (identical, all-unmasked, both chunks); the remapped restart state ($T/S/\text{hnode}$ consistent to the level, freezing cap ran: 1137 columns); the mesh files (zero elem/node level-range violations); the zero-filled dry restart levels themselves (padding them with nearest-wet values — now a permanent remap pass — reproduced the crash bitwise: the zero end-member is the boundary treatment, not the restart). The `couple_in` warning about a missing `cap_new_cavity_temp.py` is a red herring: the freezing cap runs as Fortran inside the remap binary; the warning refers to a Python duplicate that never existed.

Fix candidates (open): (i) remap-side — start draft-changed cavity columns at rest (zero u, v, w and the AB arrays at flagged columns) so the seed pump never forms and the columns adjust by gravity waves; one 4-minute test. (ii) FESOM-side, the deeper fix — a cavity-aware top boundary in the vertical tracer advection: under `ulevels>1` the top-interface exchange must carry the local concentration (material interface), not zero. Needs care to keep FCT conservation intact.

4 State of the cycle and of the climate (2026-07-17)

4.1 What is proven

- **Chunk-1 is quantitatively healthy and, under the coupled slab, crash-free:** January SST physical, zero translocated cells; OIFS Weddell September ci 0.61/0.99 (was 0.04); both-hemisphere sea-ice annual cycles; per-basin cavity melt in the observed regimes (§below); kilowatt flux events at real ice tiles: zero.
- **The workflow hands over in both directions** (Bug V fixed): `awiesm3` $\rightarrow$ PISM $\rightarrow$ `couple_out` $\rightarrow$ `chunk-3 couple_in` runs unattended.
- **The PISM decade runs on FESOM’s own melt** (`-ocean given`, 2026-07-17): real output, and a *gentle* geometry response — 290 fully-opened nodes vs 1399 in the Bug-I era, mean $|\Delta\text{draft}|$ 1.0 m, submesh 208 810 $\rightarrow$ 208 438 nodes (Fig. `pism_change_mc33`).
- **The live mesh change works:** weights regenerated and runtime-order-validated ($\leq 0.024^\circ$), restarts remapped through the `rpart` sandwich, ICMGG/`plit` regenerated.

4.2 Cavity basal melt against observations (2026-07-17)

The chunk-1 year that forced the first clean PISM decade, split by sector and set against the satellite record (Rignot et al. 2013; Adusumilli et al. 2020):

sector	observed (m ice/yr)	movcav33 chunk-1 (m w.e./yr)		
		mean	median	nodes
Filchner–Ronne	0.3–0.4	0.89	0.20	1085
Ross	~ 0.1	0.90	0.31	1075
Amundsen–Bellingshausen	14–27 (PIG 16)	7.8	6.4	229
Antarctic Peninsula	0.4–3.8	7.5	4.4	186
Wilkes (Totten sector)	4.7–9.1	8.3	6.4	296
Amery	0.6–0.7	4.0	3.3	128
E. Weddell / Dronning Maud	0.3–0.9	4.4	3.5	287
circum-Antarctic	0.8 ± 0.7	2.84	0.79	3286

The cold/warm dichotomy that dominates the observations is reproduced: the cold-cavity giants sit at 0.2–0.3 m/yr medians while the warm sectors run 4–8, and 13% of the cavity area refreezes (marine ice, as observed under Ronne/Amery). Residual biases: the East Antarctic coastal sectors (Dronning Maud, Amery) run $\sim 5\times$ warm, and the Amundsen runs $\sim 2\times$ cold — both worth revisiting once multi-year coupled statistics exist. Caveats: node statistics (unweighted), m w.e. vs the observations’ mice ($\times 0.92$), one coupled year, and PISM’s initial geometry carries reduced versions of the big shelves. For scale: the jigsaw-era coupling produced 28 m/yr.

4.3 Open item: the Southern-Hemisphere summer melt-back

The ice-thermodynamics code is byte-identical to the v3.4 reference; the difference is state and forcing. Ours keeps $\sim 2.5\times$ the reference’s winter SH extent and only a $\sim 6\times$ summer retreat where the reference manages $\sim 200\times$ (Fig. 6), and piles 2–3.5 m of snow against the Antarctic ice-shelf coast (Fig. 7). Two mitigations exist: a *clean chunk-1 ice initial state* (reference year-1849 fields indexed onto the submesh via `map_nod.out`, 7 Ronne artifact nodes hybridized; snow max 0.81 m — in use since `movcav29`), and the coupled slab itself (the pinned-skin era’s fictitious equilibria drove a spurious congelation-growth term). Whether these restore the reference’s melt-back is the

first climate question for the post-slab runs; meltponds were checked and are *not* the SH lever (the `hsno>hs1` lockout makes old and corrected schemes identical there).

5 Secondary bugs, fixed along the way

Bug	Fix	Commit
Remap extrapolated unboundedly below the old seabed (manufactured -45°C) when columns deepen	take water from the nearest old node wet at that depth	eef4f6db
Sub-ice η/hbar kept open-ocean values at newly-sub-ice nodes	zeroed (real bug, not the cure)	b17fb5c8
First mesh-change leg read the mother mesh as cavity-free (<code>cavity_nlvls.out</code> missing from the <code>mother_as_old</code> links) $\Rightarrow$ zeros in newly-wet levels, mstep-1 blowup	link the cavity files	functions
Same-chunk <code>couple_in</code> rerun promoted its own submesh to <code>previous_submesh</code> (old = new $\Rightarrow$ zeros again)	rerun guard	b9c1e370
OASIS <code>GSSPOS</code> conservation blowup on near-zero target residuals	<code>gauswgt_gsmart</code> transform	3e94d4da
<code>namcouple feom</code> dimension never actually patched (env var from another subjob's shell)	fixed variable passing	3cf29e9e
<code>fesom2ice</code> loaded the static full mesh against submesh output	load the run mesh	63cfcc91
<code>previous_submesh</code> read but never created	created; plus restart size guard	3b44a704
Failed coupling functions did not stop the workflow (backgrounded, observe saw no exit code)	<code>.coupling_failed</code> sentinel honoured by observe	7953585b
<code>rstos</code> size guard read the wrong dimension	fixed	61c8ed8a
ocp-tool flipped a coastline cell's mask+soil only, leaving an inconsistent surface column	masked nearest-neighbour rebuild of the full column	ocp-tool PR #51

The recurring pattern: the guards were right and the plumbing around them defeated them — a correct refusal followed by an unconditional continue is worse than no guard at all.

6 Falsified claims (kept, struck through)

- ~~Geostrophically-unbalanced remapped state; initialize changed cells in thermal-wind balance.~~ Failed its own offline gate (discrete ∇p noisier than NN fill); ships opt-in only, off by default.
- ~~The sub-ice ssh-BC violation is the blowup.~~ Real bug, fixed, run still died.
- ~~The per-leg PISM increment is gentle ($\approx 22\text{ m mean}$).~~ The mean was diluted by the 98% unchanged domain; the distribution is wildly skewed (max 923 m, 1399 fully-opened nodes).

4. ~~The salt-plume parameterization being off explains the warm cavity.~~ The spp-off cavity is the *coldest* in the CORE3 reference runs; the warm cavity was Bug II.
5. ~~The native ICMGG causes the mid-run crash family.~~ The step-0 swap tests proved the native, coastline-following surface set is *required* on the submesh (62 difference-zone cells cannot initialize otherwise); the mid-run family was Bug IV.
6. ~~Weighted-ist cures the crash family.~~ Necessary hygiene for the o2a blend, not sufficient: the pinned OIFS skin regenerated the instability from bounded inputs.
7. ~~The insulation/snow pile-up is the crash root.~~ It raises the loop gain (and is a real climate bias, §4), but the kW events occur on *thin* ice at 49°N; the clean-ini run crashed identically.

7 Method notes

1. **Cheap controls first.** The cold start (geometry vs state), running the fix, plotting the distribution instead of the mean — each killed a confident claim in minutes.
2. **Measurements from a corrupted run are worse than none** (Bug II poisoned Bug I’s diagnosis for a week).
3. **“Same code as the stable reference” localizes nothing by itself** — the reference ran the same FESIM solve for 500 years; the difference was the state and, one layer deeper, a switch default that pinned a skin nobody remembered was pinned.
4. **Instrument both ends of a coupling before theorizing about either.** Receive-side logging (ICEDBG) proved the flux was corrupt; send-side decomposition (SENDDBG) named the term and the tile state in its first twelve lines. The two probes cost one rebuild each.
5. **Beware silent fallbacks** (Bug V’s `isfile()` miss; the `-ocean` given silent fallback to `th`). A fallback on a nonexistent path should say so.

8 Assets

- Send/receive flux instrumentation: SENDDBG (OIFS `A_Q_ice` decomposition + tile state), ICEDBG (FESIM received flux + state at cold nodes), CPLDBG (OIFS ingest bounds) — all log-only, strip before merging.
- `oasis_expout_o2a` runscript flag: every o2a exchange dumped to netcdf ($\sim 20\times$ slowdown; crash-window probes only).
- Clean chunk-1 ice initial state (`prep/clean_ice_ini`): reference year-1849 ice fields indexed-mapped to the submesh; `map_nod.out` makes such remaps exact.
- A 2-minute offline reproducer (standalone FESOM on the PISM-cavity submesh) and a stable full-mesh control.
- `validate_feom_weights.py`: aborts any leg whose exchange weights are in the wrong node ordering — would have caught Bug II on day one.
- `verify_balance.py`: offline balance/stability gate for remapped restarts.
- Chunk-1 artifacts pooled at `input/fesom2/pism_cavity_ini/` (login-node launches).

Phantom zero-water at draft-changed cavity columns --- George VI / Marguerite Bay
black outlines: elements whose cavity draft the mesh change touched

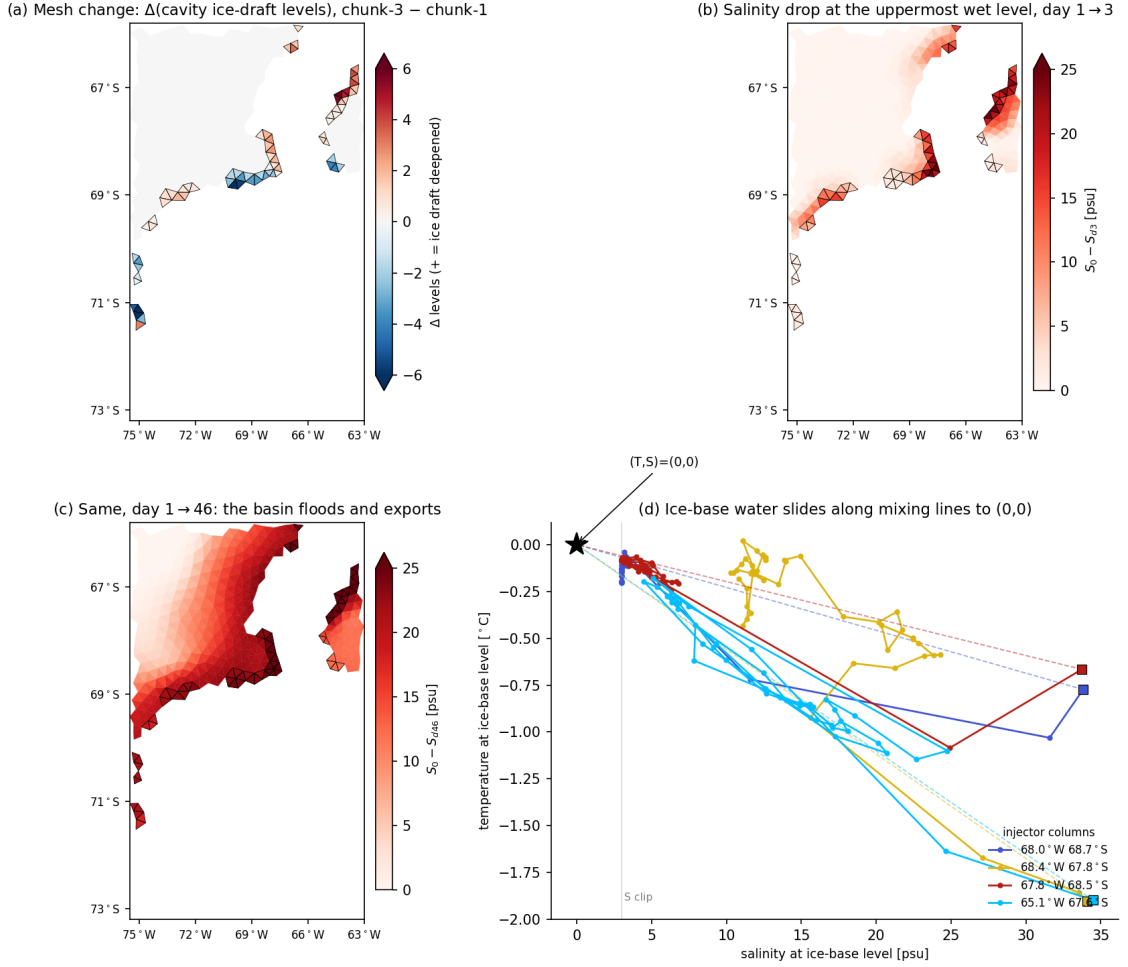


Figure 5: The chunk-3 killer, shown in one basin (George VI / Marguerite Bay; FESOM elements rendered individually, black outlines = the 73 elements whose cavity draft the mesh change touched). (a) What the mesh change did: $\pm 1-6$ cavity levels along the George VI sound and the Marguerite Bay margin. (b) By day 3 the uppermost-wet-level salinity has dropped up to 25 psu *exactly on the outlined elements* — the injection is the changed-column set. (c) By day 46 the phantom freshwater has flooded the basin and is exporting along the coast. (d) The proof: (S, T) at the ice-base level of four injector columns slides along the mixing lines (dashed) toward exactly (0,0) — T- and S-replacement fractions are identical, which only a $(T, S) = (0, 0)$ end-member produces; trajectories pin at FESOM’s $S = 3$ clip with $T \rightarrow 0$. The same pattern repeats at every fringing shelf around the continent (Getz, Victoria Land, Shackleton, West); basin-mean symptoms are the ring freshening and the coastal SSH pile-up with *negative* surface freshwater flux — the “flood” carries no water.

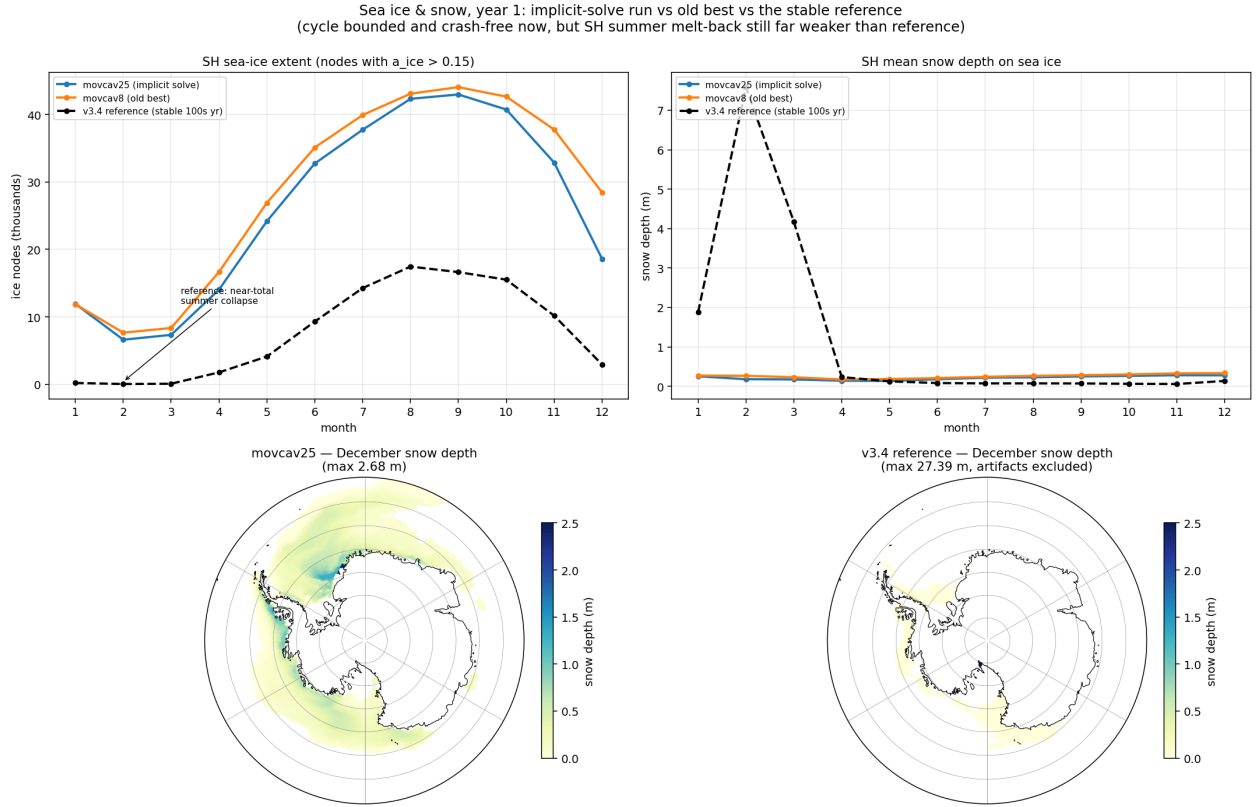


Figure 6: Year-1 sea ice and snow: implicit-solve run (blue) vs the movcav16-era state (orange) vs the 500-year-stable v3.4 reference (black dashed). Our runs track each other — the crash fixes did not degrade climate — but the reference collapses its SH ice almost completely each summer while ours retains $\sim 7k$ nodes, and its winter pack carries 0.03–0.08 m of snow against our 0.2–0.3 m. Bottom maps (same scale): December snow depth — our thick band hugs the Antarctic coast; the reference is nearly bare.

Antarctic snow-on-sea-ice pile-up at the ice-shelf coast — old movcav runs (bounded ~2-2.6 m; real ~0.5-1 m)

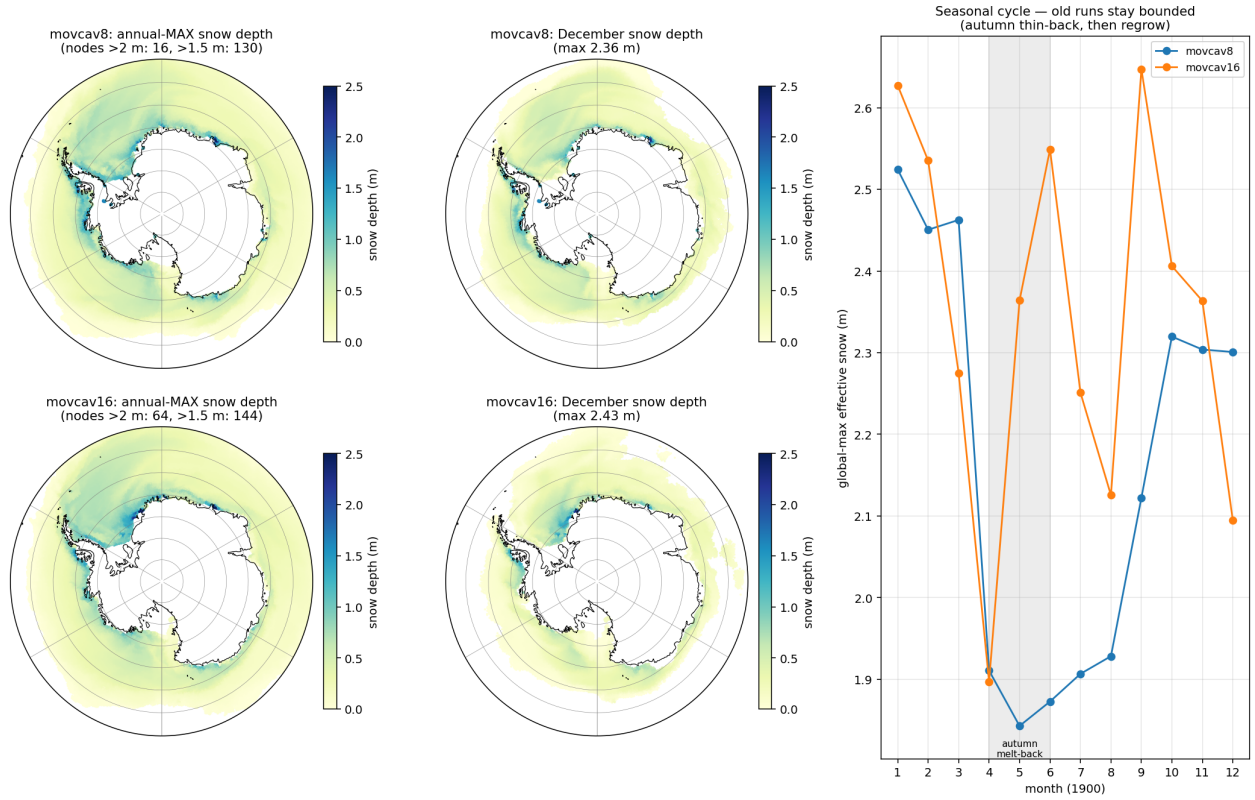


Figure 7: The extent of the coastal pile-up in the (pre-fix) movcav runs: annual-max and December snow depth, and the seasonal cycle. The thick snow is a band hugging the ice-shelf front (eastern Weddell/Fimbul, Amundsen), bounded at 2–2.6 m by an autumn melt-back. Later runs that lost this melt-back drifted monotonically to 3.5 m — the insulation that raised Bug IV’s loop gain.